Student Name: _____

Class Name: **ALEKS Integrated Mathematics II**

Number of Questions: **45**

Instructor Name: **Bodanske, Daniel**

Question 1 of 45

Solve for w.

$$3w - 8 + 2(2w + 2) = -3(w + 2)$$

Simplify your answer as much as possible.

$w =$

Question 2 of 45

Factor.

$$7v + 21$$

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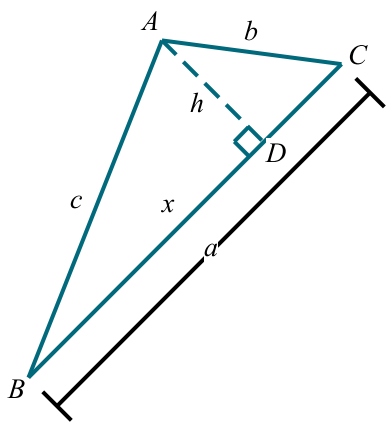
Question 3 of 45

The side lengths for $\triangle ABC$ are a, b, and c.
The height of the triangle is h, and $BD = x$.

Complete the steps below to prove the Law of Cosines.
When filling in the blanks, you may use the letters a, b, c, x, and h.



Part 1: Use the Pythagorean Theorem to find b^2.

- $b^2 = (c-x)^2 + h^2$
- $b^2 = a^2 - x^2$
- $b^2 = (a-x)^2 + h^2$
- $b^2 = x^2 + h^2$

Part 2: Use the answer from Part 1 to fill in the blanks.

$b^2 = x^2 + h^2 + __^2 - 2 \cdot __ \cdot __$

Part 3: Use the Pythagorean Theorem to find c^2.

- $c^2 = (b-x)^2 + h^2$
- $c^2 = (a-x)^2 + h^2$
- $c^2 = a^2 - b^2$
- $c^2 = x^2 + h^2$

Part 4: Use the answers from Parts 2 and 3 to fill in the blanks.

$b^2 = __^2 + __^2 - 2 \cdot __ \cdot __$

Part 5: Use trigonometry to fill in the blank.

$x = __ \cos B$

Part 6: Use the answers from Parts 4 and 5 to fill in the blanks.

$b^2 = __^2 + __^2 - 2 \cdot __ \cdot __ \cos B$

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Question 4 of 45

Fill in the blank with a number to make the expression a perfect square.

$$v^2 + 10v + \square$$

$$v^2 + 10v + \square$$

Question 5 of 45

The equation of a circle is given below. Identify the radius and the center. Then graph the circle.

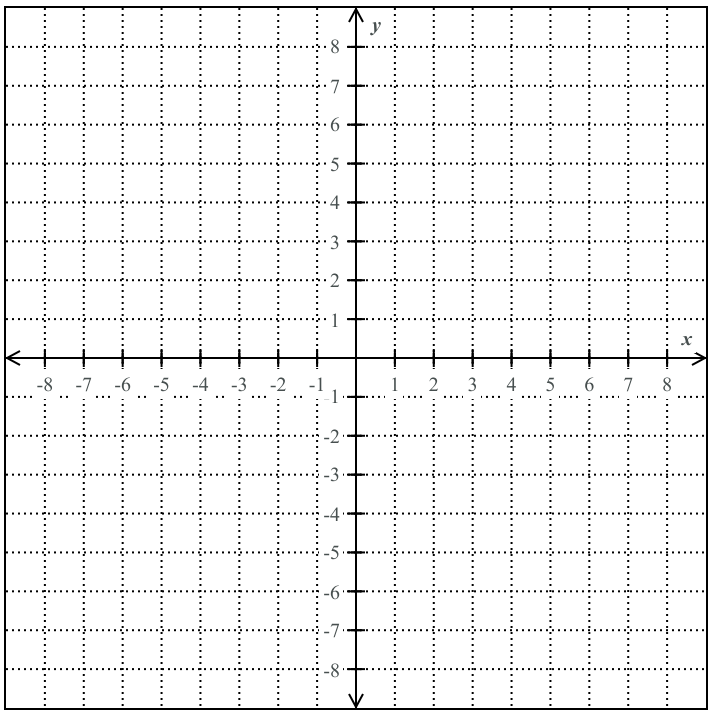
$$x^2 + y^2 - 6x + 4y - 3 = 0$$

(a) Identify the center and the radius of the circle.

Radius: _____

Center: (____ , ____)

(b) Graph the circle.



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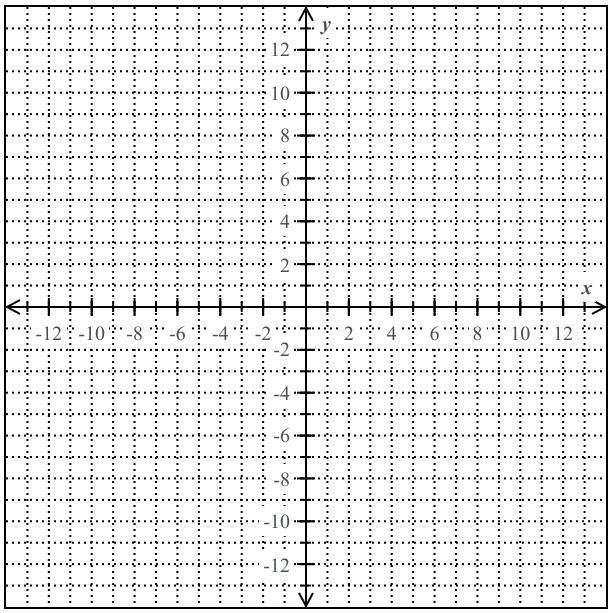
Question 6 of 45

Graph the parabola.

$$y = 3x^2 - 24x + 43$$



Question 7 of 45

What are the degree and leading coefficient of the polynomial?

$$-10v - 8v^2 - 5v^6 + 9$$

Degree:

Leading coefficient:

Question 8 of 45

Simplify.

$$\left(9u^2 + 2u + 3\right) + \left(5u^2 - 4u - 9\right) - \left(-2u^2 + 7u + 1\right)$$

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Question 9 of 45

Simplify.

$$\left(6x^3y^2 + 5x^2y^3 - 7y^2\right) + \left(-x^2y^3 - 3x^3y^2 + 2y^2\right)$$

Question 10 of 45

For each statement, choose whether it is *always*, *sometimes*, or *never* true.

An integer multiplied by an integer is an integer.

_ Always true
_ Sometimes true
_ Never true

An integer added to an integer is an integer.

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Question 11 of 45

Solve for x.

$$6(x-5)-4=-6(-3x+5)-2x$$

Simplify your answer as much as possible.

$x =$ ☐

Question 12 of 45

Use the distributive property to remove the parentheses.

$$\left(9z^4-7z^2+4\right)10z^5$$

Simplify your answer as much as possible.

Question 13 of 45

Rewrite without parentheses.

$$-6w^2\left(3w^3-9w^2+6w\right)$$

Simplify your answer as much as possible.

Question 14 of 45

Rewrite without parentheses.

$$-6b^5c\left(9b^6+7b^2c^3-5c^3\right)$$

Simplify your answer as much as possible.

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Question 15 of 45

For each statement, choose whether it is *always*, *sometimes*, or *never* true.

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Question 16 of 45

Multiply.

$$(a+4b)(8a-3b)$$

Simplify your answer.

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Question 17 of 45

Multiply.

$$(-5c+4)(-8c+5)$$

Simplify your answer.

Question 18 of 45

Multiply.

$$\left(4v^2+v+7\right)\left(6v^2-6v+3\right)$$

Simplify your answer.

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Question 19 of 45

For each statement, choose whether it is *always*, *sometimes*, or *never* true.

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Question 20 of 45

Multiply.

$$(9y - 8x)(9y + 8x)$$

Simplify your answer.

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Question 21 of 45

Rewrite without parentheses and simplify.

$$(5u+4x)^2$$

Question 22 of 45

Factor.

$$3w-18$$

Question 23 of 45

Find the greatest common factor of $6a^2$ and $15n^4$.

Question 24 of 45

Factor.

$$15y-11y^2$$

Question 25 of 45

Factor the following expression.

$$30v^4w^5y^2-24v^7w^7$$

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Question 26 of 45

Factor by grouping.

$$4u^3 + 7u^2 + 24u + 42$$

Question 27 of 45

Factor by grouping.

$$x^6 - 6x^5 + 5x - 30$$

Question 28 of 45

Factor by grouping.

$$5p - 3ps + 5m - 3ms$$

Question 29 of 45

Factor.

$$x^2 + 16x + 39$$

Question 30 of 45

Factor.

$$x^2 + 3xy - 4y^2$$

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Question 31 of 45

Factor completely.

$$3u^2 - 42u + 72$$

Question 32 of 45

Factor.

$$5y^2 - 14y + 8$$

Question 33 of 45

Factor.

$$3z^2 - 19z + 30$$

Question 34 of 45

Factor.

$$18y^2 + 27y + 10$$

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Question 35 of 45

Factor by grouping (sometimes called the *ac*-method).

$$6x^2 + 13x - 5$$

First, choose a form with appropriate signs.
Then, fill in the blanks with numbers to be used for grouping.
Finally, show the factorization.

Form:

- ○ $6x^2 + ____\, x + ____\, x - 5$
- ○ $6x^2 + ____\, x - ____\, x - 5$
- ○ $6x^2 - ____\, x + ____\, x - 5$
- ○ $6x^2 - ____\, x - ____\, x - 5$

Factorization:

_____

Question 36 of 45

Factor completely.

$$-3u^2 - 27u - 60$$

Question 37 of 45

Factor.

$$u^2 + 10u + 25$$

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Question 38 of 45

Factor.

$$9w^2 - 24w + 16$$

Question 39 of 45

Factor.

$$36 - v^2$$

Question 40 of 45

Factor.

$$25x^2 - 64$$

Question 41 of 45

Factor.

$$81z^2 - x^2$$

Question 42 of 45

Factor completely.

$$32y^4 - 50y^2$$

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Question 43 of 45

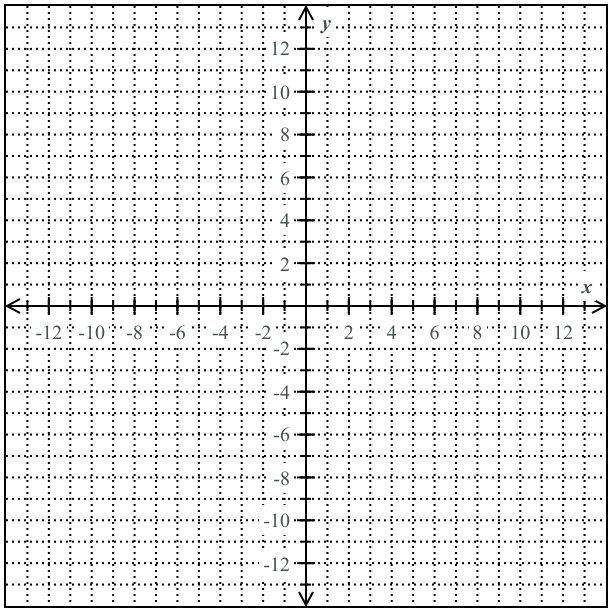
Factor completely.

$$3u^4 - 6u^3 + 33u^2$$

Question 44 of 45

Graph the parabola.

$$y = 2x^2 + 8x + 10$$



Question 45 of 45

Fill in the blank with a number to make the expression a perfect square.

$$v^2 + 6v + \square$$

$$v^2 + 6v + \square$$

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Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **45**

Question 1 of 45

$$w = -\frac{1}{5}$$

Question 2 of 45

$$7(v+3)$$

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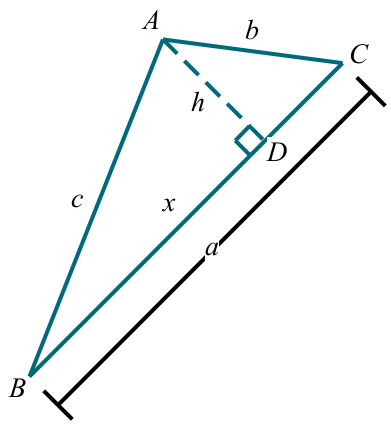
Question 3 of 45



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- ○ $b^2 = a^2 - x^2$
- ◉ $b^2 = (a-x)^2 + h^2$
- ○ $b^2 = x^2 + h^2$

Part 2: Use the answer from Part 1 to fill in the blanks.

$b^2 = x^2 + h^2 + a^2 - 2 \cdot a \cdot x$

Part 3: Use the Pythagorean Theorem to find c^2.

- ○ $c^2 = (b-x)^2 + h^2$
- ○ $c^2 = (a-x)^2 + h^2$
- ○ $c^2 = a^2 - b^2$
- ◉ $c^2 = x^2 + h^2$

Part 4: Use the answers from Parts 2 and 3 to fill in the blanks.

$b^2 = c^2 + a^2 - 2 \cdot a \cdot x$

Part 5: Use trigonometry to fill in the blank.

$x = c \cdot \cos B$

Part 6: Use the answers from Parts 4 and 5 to fill in the blanks.

$b^2 = c^2 + a^2 - 2 \cdot a \cdot c \cdot \cos B$

Question 4 of 45

$v^2 + 10v + 25$

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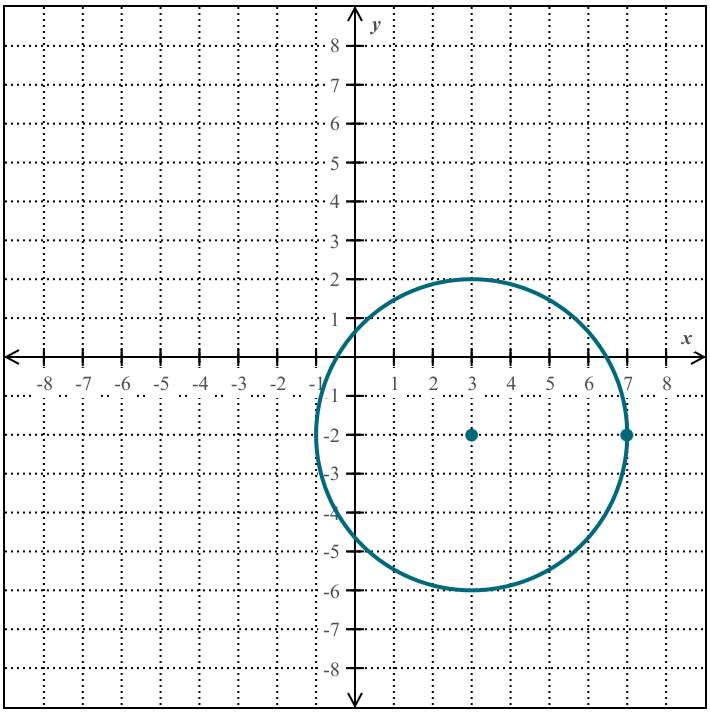
Question 5 of 45

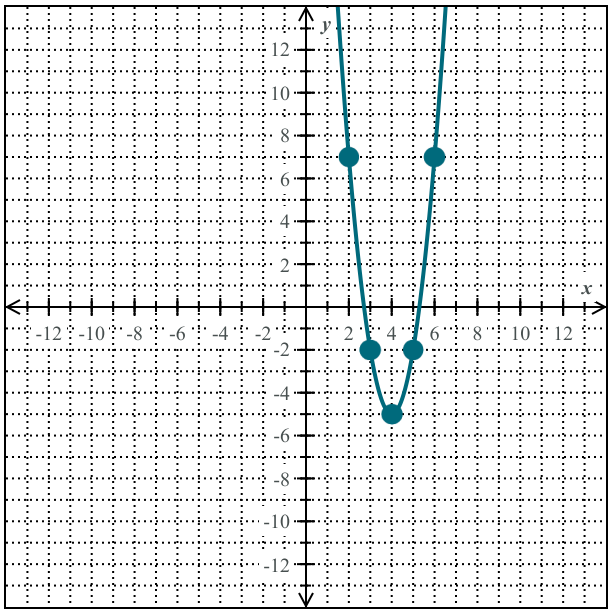
(a) Radius: 4

Center: $(3, -2)$

(b)



Question 6 of 45



Question 7 of 45

Degree: 6

Leading coefficient: -5

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Question 8 of 45

$16u^2 - 9u - 7$

Question 9 of 45

$3x^3y^2 + 4x^2y^3 - 5y^2$

Question 10 of 45

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Question 11 of 45

$x = -\frac{2}{5}$

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Question 12 of 45

$90z^9 - 70z^7 + 40z^5$

Question 13 of 45

$-18w^5 + 54w^4 - 36w^3$

Question 14 of 45

$-54b^{11}c - 42b^7c^4 + 30b^5c^4$

Question 15 of 45

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Question 16 of 45

$8a^2 + 29ab - 12b^2$

Question 17 of 45

$40c^2 - 57c + 20$

Question 18 of 45

$24v^4 - 18v^3 + 48v^2 - 39v + 21$

Question 19 of 45

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Question 20 of 45

$81y^2 - 64x^2$

Question 21 of 45

$25u^2 + 40ux + 16x^2$

Question 22 of 45

$3(w-6)$

Question 23 of 45

3

Question 24 of 45

$y(15-11y)$

Question 25 of 45

$6v^4 w^5 \left(5y^2 - 4v^3 w^2\right)$

Question 26 of 45

$(4u+7)\left(u^2+6\right)$

Question 27 of 45

$(x-6)\left(x^5+5\right)$

Question 28 of 45

$(5-3s)(p+m)$

Question 29 of 45

$(x+13)(x+3)$

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Question 30 of 45

$(x-y)(x+4y)$

Question 31 of 45

$3(u-12)(u-2)$

Question 32 of 45

$(5y-4)(y-2)$

Question 33 of 45

$(3z-10)(z-3)$

Question 34 of 45

$(6y+5)(3y+2)$

Question 35 of 45

Form:

- ○ $6x^2 + ____ x + ____ x - 5$
- ○ $6x^2 + ____ x - ____ x - 5$
- ◉ $6x^2 - 2x + 15x - 5$
- ○ $6x^2 - ____ x - ____ x - 5$

Factorization:

$(3x-1)(2x+5)$

Question 36 of 45

$-3(u+4)(u+5)$

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Question 37 of 45

$(u+5)^2$

Question 38 of 45

$(3w-4)^2$

Question 39 of 45

$(6+v)(6-v)$

Question 40 of 45

$(5x+8)(5x-8)$

Question 41 of 45

$(9z+x)(9z-x)$

Question 42 of 45

$2y^2(4y+5)(4y-5)$

Question 43 of 45

$3u^2\left(u^2-2u+11\right)$

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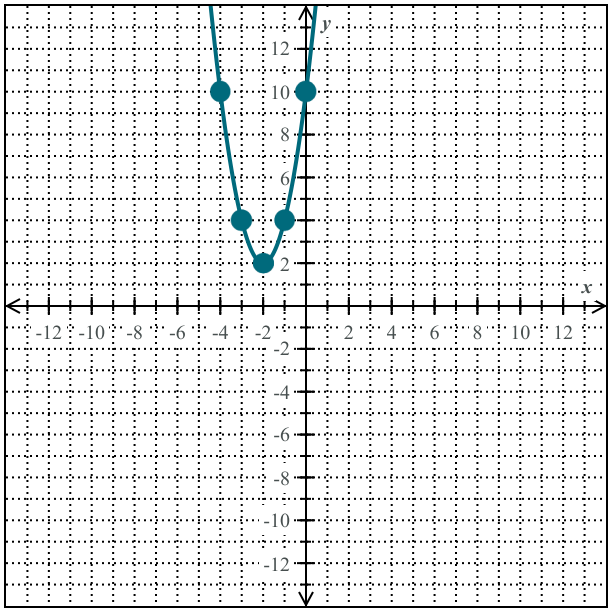
Question 44 of 45



Question 45 of 45

$v^2 + 6v + 9$

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